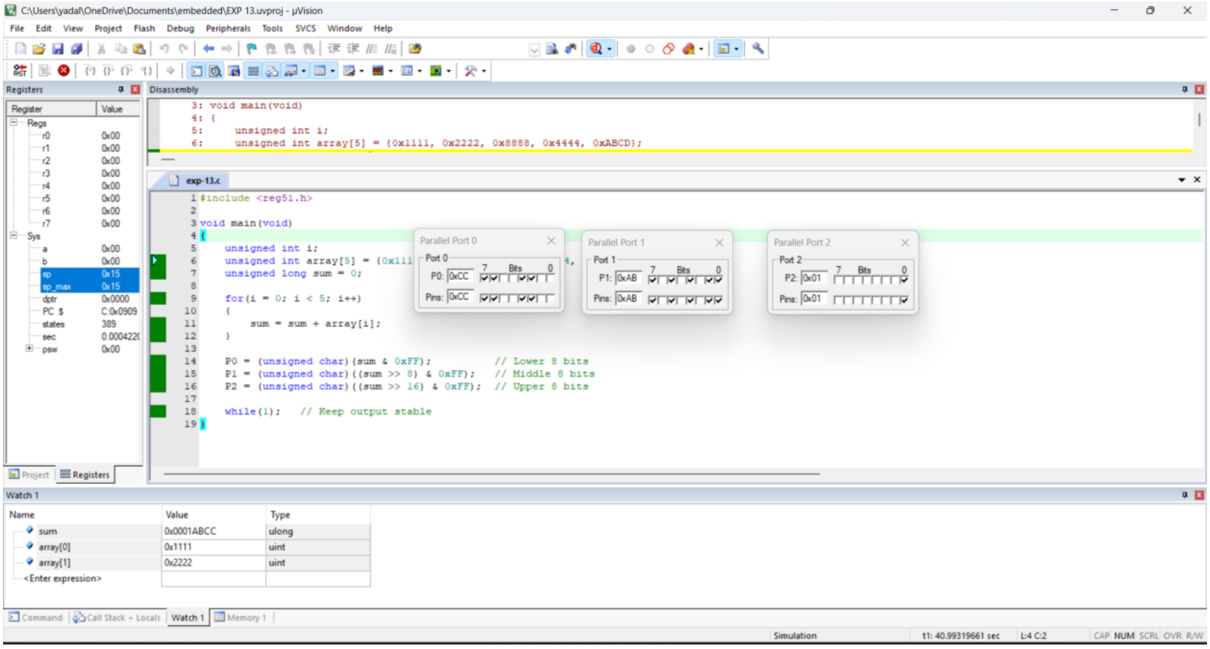
Exp No: 13 EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

Github : https://github.com/Pardhav2006/embedded-system

Program:

```
#include <reg51.h>

void main(void)

{

   unsigned int i;

   unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};

   unsigned long sum = 0;

   for(i = 0; i < 5; i++)

   {

      sum = sum + array[i];

   }

   P0 = (unsigned char)(sum & 0xFF);         // Lower 8 bits

   P1 = (unsigned char)((sum >> 8) & 0xFF);   // Middle 8 bits

   P2 = (unsigned char)((sum >> 16) & 0xFF);  // Upper 8 bits

   while(1);   // Keep output stable
```

}


C:\Users\yadal\OneDrive\Documents\embedded\EXP 13.uvproj - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Disassembly
3: void main(void)
4: {
5: unsigned int i;
6: unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
exp-13.c
#include <reg51.h>
void main(void)
{
unsigned int i;
unsigned long sum = 0;
for(i = 0; i < 5; i++)
{
sum = sum + array[i];
}
P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
P1 = (unsigned char)((sum >> 8) & 0xFF); // Middle 8 bits
P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
while(1); // Keep output stable
}
Parallel Port 0
P0: 0xCC
Pins: 0xCC
Parallel Port 1
P1: 0xAB
Pins: 0xAB
Parallel Port 2
P2: 0x01
Pins: 0x01
Watch 1
Name Value Type
sum 0x0001ABCC ulong
array[0] 0x1111 uint
array[1] 0x2222 uint
<Enter expression>
Simulation
t1: 40.99319661 sec